

C04 -AT3
MLA 0201 Fundamentals Of Machine Learning

Annexure A

Comparative Analysis

question 1 (20 Marks)

A healthcare organization is developing an intelligent disease diagnosis system using patient symptoms and medical history.

Compare and analyze the suitability of **Bayesian Networks** and **Markov Random Fields (MRFs)** for modeling the relationships among diseases and symptoms. Compare both models with respect to graph structure, conditional independence assumptions, inference process, uncertainty handling, computational complexity, and real-world applicability. Justify which model is more appropriate for this healthcare application.

Ans:Comparative Analysis: Bayesian Networks vs Markov Random

Fields 1. Graph Structure

- **Bayesian Network (BN):** Uses a **Directed Acyclic Graph (DAG)**. Arrows represent cause-and-effect relationships, such as **Disease → Symptoms**.
- **Markov Random Field (MRF):** Uses an **undirected graph**, where connections represent relationships between diseases and symptoms without a direction.

2. Conditional Independence

- **BN:** Represents conditional independence using the direction of relationships and dependencies between parent and child nodes.
- **MRF:** Represents conditional independence using the concept of **graph separation** and neighboring nodes.

3. Inference Process

- **BN:** Uses probabilistic inference to calculate the probability of a disease based on observed symptoms and medical history.

4. Handling Uncertainty

- **BN:** Handles uncertainty very effectively and can combine symptoms, medical history, and prior disease probabilities.
- **MRF:** Also handles uncertainty well, but it is generally more suitable when relationships are symmetric rather than causal.

5. Computational Complexity

- **BN:** Inference can become computationally expensive for large and highly connected networks, but many healthcare models can be designed efficiently.
- **MRF:** Inference can also be computationally expensive, particularly for large graphs with many interconnected variables.

6. Justification

For this **disease diagnosis application**, **Bayesian Networks are more appropriate**.

The main reason is that healthcare diagnosis naturally involves **cause-and-effect relationships**. For example:

Disease → Symptoms → Test Results

A Bayesian Network can calculate the probability of a disease when symptoms and medical history are provided. It can also incorporate prior medical knowledge and handle missing or uncertain information.

Conclusion

Although both Bayesian Networks and MRFs are useful probabilistic models, **Bayesian Networks are better suited for an intelligent disease diagnosis system** because they clearly represent causal relationships, provide interpretable probability-based predictions, and effectively handle uncertainty in patient symptoms and medical history.

Question 2 (20 Marks)

An autonomous vehicle company is designing a system to recognize driving activities from continuous sensor data collected over time.

Compare and evaluate the performance of **Hidden Markov Models (HMMs)** and **Conditional Random Fields (CRFs)** for sequence labeling and activity recognition. Compare the models based on state representation, observation modeling, learning approach, prediction accuracy, handling of contextual features, computational efficiency, and scalability. Recommend the most suitable model for the given application with proper justification.

Ans: Comparative Analysis: HMM vs CRF for Autonomous Vehicle Activity

Recognition

1. State Representation

- **HMM:** Represents driving activities as **hidden states**, such as *Driving*, *Braking*, *Turning*, and *Stopping*. Sensor readings are treated as observations.
- **CRF:** Directly models the relationship between the **observed sensor data and activity labels**.

2. Observation Modeling

- **HMM:** Uses **emission probabilities** to determine how likely an observation is for a particular hidden state. It commonly assumes observations are conditionally independent given the state.
- **CRF:** Can use many overlapping features from the current and neighboring observations without requiring the same independence assumptions.

3. Learning Approach

- **HMM:** Learns **transition probabilities** between states and **emission probabilities** for observations.
- **CRF:** Learns feature weights from labeled training sequences to determine the most likely sequence of activity labels.

4. Prediction Accuracy

- **HMM:** Performs well when the data follows its probabilistic assumptions and activities have clear transitions.
- **CRF:** Often provides better accuracy for complex sensor data because it can consider multiple contextual features simultaneously.

5. Contextual Features

- **HMM:** Mainly considers the current hidden state and previous state.
- **CRF:** Can use rich contextual information, such as **current speed, previous speed, acceleration, steering angle, and neighboring sensor readings**.

6. Computational Efficiency

- **HMM:** Generally simpler and computationally efficient, especially for smaller sequence problems. The **Viterbi algorithm** can efficiently find the most likely state sequence.
- **CRF:** Training can be more computationally expensive because it learns many feature weights, although inference can still be performed efficiently using dynamic programming.

Question 3 (20 Marks)

A financial analytics company is building an AI-based risk prediction system using customer transaction data and historical financial records.

Compare Directed Graphical Models (Bayesian Networks) and Undirected Graphical Models (Markov Random Fields) for representing probabilistic dependencies in the system. Evaluate both approaches in terms of graph representation, dependency modeling, inference techniques, learning algorithms, generalization capability, computational requirements, and suitability for financial risk prediction. Conclude which graphical model is more effective for this industrial application and justify your recommendation.

Ans:Comparative Analysis: Bayesian Networks vs Markov Random Fields for Financial Risk Prediction

1. Graph Representation

- **Bayesian Network (BN):** Uses a **directed acyclic graph (DAG)**. Arrows show the direction of dependency between variables.
- **Markov Random Field (MRF):** Uses an **undirected graph**, where connections represent relationships between variables without a direction.

2. Dependency Modeling

- **BN:** Clearly represents causal or directional relationships, such as **Income → Loan Approval → Default Risk**.
- **MRF:** Represents mutual relationships between variables and is useful when dependencies are symmetric.

3. Inference Techniques

- **BN:** Uses methods such as **Bayesian inference, variable elimination, and belief propagation** to calculate probabilities.
- **MRF:** Commonly uses **belief propagation, Gibbs sampling, and other probabilistic inference methods**.

4. Learning Algorithms

- **BN:** Can learn network structure and probability parameters from historical financial data.
- **MRF:** Learns parameters such as **weights/potentials** that represent relationships between connected variables.

5. Generalization Capability

- **BN:** Can generalize well by using prior probabilities and relationships between variables. It is particularly useful when the causal structure is known.
- **MRF:** Can generalize well when variables have strong interconnected relationships, but interpretation can be more difficult.

6. Computational Requirements

- **BN:** Inference can become expensive for very large and highly connected networks, but structured networks can be relatively efficient.
- **MRF:** Inference can also be computationally expensive, especially when the graph contains many interconnected variables and complex dependencies.

7.Recommendation

For **financial risk prediction**, **Bayesian Networks are generally more effective**. Financial risk analysis often involves directional relationships between variables. For example:

Customer Income → Credit Score → Loan Approval → Default Risk

A Bayesian Network can represent these relationships and calculate the probability of events such as **loan default or financial risk** based on customer transaction history and financial records.

It also provides interpretable results, which are important in financial applications where organizations need to understand **why a customer was classified as high-risk**.

Conclusion

Both models can represent uncertainty and dependencies effectively. **MRFs are useful for complex, interconnected relationships**, while **Bayesian Networks are better when directional relationships and interpretability are important**. Therefore, **Bayesian Networks are recommended for this financial risk prediction system** because they provide clear dependency representation, probabilistic reasoning, and interpretable risk predictions.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
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Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to

					understand
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